Stanley Avril

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Education

Polytechnique Montréal Expected fall 2027
B.Eng., Software Engineering Montréal, QC

Experience

Jaunt Air Mobility May 2026 to Present
Software Engineering Intern: Flight Software and RPAS Communications Montréal, QC

- Built a Python/Tkinter dashboard so ground control operators could see connection quality and outages across LTE, ZeroTier VPN, SATCOM, and Starlink. Added PNG and HTML reports for reviewing telemetry after a flight.
- Split a 1,500+ line Python ground control application into modules so individual features could be changed and tested without working through one large file.
- Wrote a Python service on Raspberry Pi to send system health and connection status to the ground station every 10 seconds using MAVLink.
- Connected a Thales GATEWAY satellite uplink to the aircraft position feed using UDP and REST APIs so ground control could receive positions through the satellite link.

Propolys May to Aug 2025
Cohort Participant: Startup Accelerator Montréal, QC

- Interviewed 50+ users to decide what to build in an MVP. Built login, API, and dashboard features, then discussed progress with mentors and presented the demo.

Le Polyscope May to Aug 2024
Web Developer: Polytechnique Student Newspaper Montréal, QC

- Updated the student newspaper website using React, JavaScript, HTML, CSS, and Firebase. Fixed layouts and interface issues so articles were easier to read on phones and computers.
- Worked with the team in Agile sprints, using Git and GitLab merge requests to review fixes before merging them into the website.

Selected Projects

CodeCell Live: Collaborative Browser IDE Jul 2025 to Present
React / TypeScript / Monaco / tRPC / WebSocket / Go / Firecracker Code

- Built a browser editor with React, TypeScript, and Monaco so users could work on the same code. Used WebSocket to share edits, cursor positions, and program output between clients.
- Used Go and Firecracker to run Python in separate virtual machines so code runs stayed isolated. Added time and memory limits and cleaned up temporary files after execution.

SpaceMind: Local Storage Assistant Jul 2026 to Present
Rust / SQLite / BLAKE3 / Ollama Code

- Wrote a Rust file scanner with BLAKE3 duplicate detection so users could see what was taking up disk space and review files before cleaning up.
- Used SQL and SQLite to save scan history and review decisions. Added Ollama explanations from file metadata and file checks before confirmed Trash actions so users could review suggestions before changing files.

SMS Booking AI Agent Dec 2025 to Jun 2026
TypeScript / NestJS / Twilio / OpenAI / Google Calendar / Supabase Demo

- Built a TypeScript/NestJS backend using Twilio and OpenAI so customers could request appointments by SMS. The agent asks for the service, address, and time, then requests confirmation.
- Used the Google Calendar REST API to check availability and add confirmed appointments. Saved conversation state in Supabase and tested the booking flow with Jest and real phone messages.

Skills

Languages: Python, TypeScript, C/C++, Go, Rust, JavaScript, SQL
Web & Data: React, HTML, CSS, NestJS, Node.js, WebSocket, tRPC, SQLite, Supabase, OpenAI API
Systems & Tools: MAVLink, ZeroTier VPN, SATCOM, Raspberry Pi, Git, Docker, GitLab CI/CD, Jest
Spoken languages: French (native), English (fluent)